

Project II: Sliding Tiles

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Directory Hierarchy

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Namespace Index

2.1 Namespace List

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Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

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Chapter 5

Directory Documentation

5.1 Project II Slider Game Directory Reference

Files

- file [Board.cpp](#)
File containing the initialization of all [Board](#) methods.
- file [Board.hpp](#)
File containing a [Board](#) class.
- file [Main.cpp](#)
Main file where the program is executed.
- file [SlidingTilesData.cpp](#)
File containing sliding tiles variable initializations.
- file [SlidingTilesData.hpp](#)
File containing a namespace data for a Sliding Tiles game.
- file [SlidingTilesEnums.hpp](#)
File containing a namespace with enum classes for a Sliding Tiles game.
- file [SlidingTilesFunctions.cpp](#)
File containing the initialization of the [SlidingTilesFunctions](#) functions.
- file [SlidingTilesFunctions.hpp](#)
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- file [SlidingTilesFunctionsHelpers.cpp](#)
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File containing a namespace of declared helper functions and a constant expression for [SlidingTilesFunctions](#).
- file [UserFunctions.cpp](#)
File containing the initialization of the [UserFunctions](#) functions.
- file [UserFunctions.hpp](#)
File containing a namespace with declared functions that the user interacts with.
- file [UserFunctionsHelpers.cpp](#)
File containing the initialization of the [UserFunctionsHelpers](#) functions.
- file [UserFunctionsHelpers.hpp](#)
File containing a namespace of declared helper functions and a constant expression for [UserFunctions](#).

Chapter 6

Namespace Documentation

6.1 SlidingTilesData Namespace Reference

Namespace containing data for a Sliding Tiles game.

Variables

- [Board board](#)
The [Board](#) object for Sliding Tiles.
- unsigned int [slides](#) = 0
The amount of slides performed.
- size_t [currentRow](#) = 0
The current row index of the empty tile.
- size_t [currentColumn](#) = 0
The current column index of the empty tile.
- size_t [boardSize](#) = 0
The size of the board in a single direction.
- [SlidingTilesEnums::Difficulty currentDifficulty](#) = [SlidingTilesEnums::Difficulty::EASY](#)
The current difficulty being played.

6.1.1 Detailed Description

Namespace containing data that changes and persists between functions during a game of Sliding Tiles.

Author

Rajdeep Chowdhury

Date

8.04.2026

6.1.2 Variable Documentation

6.1.2.1 board

```
Board SlidingTilesData::board
```

The `Board` object which is used during a Sliding Tiles game.

Definition at line 20 of file [SlidingTilesData.cpp](#).

6.1.2.2 boardSize

```
size_t SlidingTilesData::boardSize = 0
```

The amount of tiles the board has in a single direction.

Definition at line 23 of file [SlidingTilesData.cpp](#).

6.1.2.3 currentColumn

```
size_t SlidingTilesData::currentColumn = 0
```

The column index of the empty tile of the Sliding Tiles game.

Definition at line 22 of file [SlidingTilesData.cpp](#).

6.1.2.4 currentDifficulty

```
SlidingTilesEnums::Difficulty SlidingTilesData::currentDifficulty = SlidingTilesEnums::Difficulty::EASY
```

The difficulty of the Sliding Tiles game being played.

Definition at line 24 of file [SlidingTilesData.cpp](#).

6.1.2.5 currentRow

```
size_t SlidingTilesData::currentRow = 0
```

The row index of the empty tile of the Sliding Tiles game.

Definition at line 22 of file [SlidingTilesData.cpp](#).

6.1.2.6 slides

```
unsigned int SlidingTilesData::slides = 0
```

The amount of slides which occur during a Sliding Tiles game.

Definition at line 21 of file [SlidingTilesData.cpp](#).

6.2 SlidingTilesEnums Namespace Reference

Namespace containing enum classes for a Sliding Tiles game.

Enumerations

- enum class [Difficulty](#) { [EASY](#) = 3 , [MEDIUM](#) = 4 , [HARD](#) = 6 , [INSANE](#) = 9 }
Enum class containing difficulties for a Sliding Tiles game.
- enum class [Direction](#) { [UP](#) , [DOWN](#) , [LEFT](#) , [RIGHT](#) }
Enum class containing directions that a slide can occur.

6.2.1 Detailed Description

Namespace containing enum classes that represent the difficulty of a Sliding Tiles game and the directions which a slide can occur.

Author

Rajdeep Chowdhury

Date

8.04.2026

6.2.2 Enumeration Type Documentation

6.2.2.1 Difficulty

```
enum class SlidingTilesEnums::Difficulty [strong]
```

Enum class containing difficulties for a Sliding Tiles game. The integers of each enum represent the size of board they produce.

Author

Rajdeep Chowdhury

Date

8.04.2026

Enumerator

EASY	EASY Difficulty . Has a board size of 3. The easy difficulty for Sliding Tiles. Has a board size of 3, meaning a 3x3 board is produced.
------	---

MEDIUM	MEDIUM Difficulty . Has a board size of 4. The medium difficulty for Sliding Tiles. Has a board size of 4, meaning a 4x4 board is produced.
HARD	HARD Difficulty . Has a board size of 6. The hard difficulty for Sliding Tiles. Has a board size of 6, meaning a 6x6 board is produced.
INSANE	INSANE Difficulty . Has a board size of 9. The insane difficulty for Sliding Tiles. Has a board size of 9, meaning a 9x9 board is produced.

Definition at line 24 of file [SlidingTilesEnums.hpp](#).

6.2.2.2 Direction

```
enum class SlidingTilesEnums::Direction [strong]
```

Enum class containing the four directions that a slide can be performed in. It is used in the context of the empty tile sliding to an adjacent filled tile, rather than the vice versa.

Author

Rajdeep Chowdhury

Date

8.04.2026

Enumerator

UP	UP Direction . Enum used to slide the empty tile up.
DOWN	DOWN Direction . Enum used to slide the empty tile down.
LEFT	LEFT Direction . Enum used to slide the empty tile left.
RIGHT	RIGHT Direction . Enum used to slide the empty tile right.

Definition at line 51 of file [SlidingTilesEnums.hpp](#).

6.3 SlidingTilesFunctions Namespace Reference

Namespace containing the functions used during a game of Sliding Tiles.

Functions

- void [startGame](#) ([SlidingTilesEnums::Difficulty](#) difficulty)
Sets up the data required to start a Sliding Tiles game.
- bool [slide](#) ([SlidingTilesEnums::Direction](#) direction)
Attempts to slide the empty tile to an adjacent filled tile.
- void [shuffle](#) ()
Shuffles the board.
- bool [isBoardOrdered](#) ()
Checks if all the board's numbers are ordered.
- void [writeScore](#) ([SlidingTilesEnums::Difficulty](#) difficulty)
Writes the number of slides to its associated leaderboard.
- std::vector< unsigned int > [loadScores](#) ([SlidingTilesEnums::Difficulty](#) difficulty)
Loads the top scores from its associated leaderboard.

6.3.1 Detailed Description

Namespace containing the functions which are called during a game of Sliding Tiles to change the properties of [SlidingTilesData](#).

Author

Rajdeep Chowdhury

Date

8.04.2026

See also

[SlidingTilesData](#)

6.3.2 Function Documentation

6.3.2.1 isBoardOrdered()

```
bool SlidingTilesFunctions::isBoardOrdered ()
```

Checks if all the board's numbers are in their expected, ordered position, reading the board from left to right and from top to bottom.

Return values

<i>true</i>	If the board is ordered.
<i>false</i>	If the board is not ordered.

Author

Fabio Bustamante Romillo

Date

8.04.2026

Definition at line 114 of file [SlidingTilesFunctions.cpp](#).

6.3.2.2 loadScores()

```
std::vector< unsigned int > SlidingTilesFunctions::loadScores (
    SlidingTilesEnums::Difficulty difficulty)
```

Loads all scores from a text file representing the leaderboard of the given `difficulty` into a vector, sorts them in ascending order, then filters the scores to the top [SlidingTilesFunctionsHelpers::LEADERBOARD_SIZE](#) scores.

Parameters

in	<i>difficulty</i>	The difficulty of the Sliding Tiles game leaderboard.
----	-------------------	---

Returns

A vector of the top scores.

Author

Raj Bahadur Bhat

Date

8.04.2026

See also

[SlidingTilesFunctionsHelpers::LEADERBOARD_SIZE](#)

Definition at line 155 of file [SlidingTilesFunctions.cpp](#).

6.3.2.3 shuffle()

```
void SlidingTilesFunctions::shuffle ()
```

Shuffles [SlidingTilesData::board](#) by repeatedly calling [slide\(SlidingTilesEnums::Direction\)](#) with a random direction. The shuffle will not immediately backtrack, meaning it can't slide one direction and then slide back the opposite direction immediately afterwards.

Precondition

[SlidingTilesData::board](#) is not empty and all rows contain at least one element. (i.e., the [Board::hasEmptyRow\(\)](#) call returns false.)

Author

Fabio Bustamante Romillo

Date

8.04.2026

Note

If the pre-condition is not met, the function returns early without modifying the data.

The function `SlidingTilesFunctions::StartGame(SlidingTilesEnums::Difficulty)` should be used before calling this function.

Warning

If either one of the dimensions of the `SlidingTilesData::board` is somehow equal to 1, this function will enter an infinite loop due to its incapability to backtrack. This should be impossible at the moment, but it is worth noting if this code is to be expanded on.

See also

[SlidingTilesFunctionsHelpers](#)

[Board::hasEmptyRow\(\)](#)

Definition at line 87 of file [SlidingTilesFunctions.cpp](#).

6.3.2.4 slide()

```
bool SlidingTilesFunctions::slide (
    SlidingTilesEnums::Direction direction)
```

Attempts to swap the empty tile with an adjacent filled tile based on the `direction` given. A slide will fail if the direction of slide given would cause the empty tile to slide outside the board.

Precondition

`SlidingTilesData::board` is not empty and all rows contain at least one element. (i.e., the [Board::hasEmptyRow\(\)](#) call returns false.)

Parameters

<i>in</i>	<i>direction</i>	The direction that the slide will occur.
-----------	------------------	--

Return values

<i>true</i>	If the slide succeeds.
<i>false</i>	If the slide fails.
<i>false</i>	If the pre-condition is not met.

Author

Rajdeep Chowdhury

Date

8.04.2026

Note

If the pre-condition is not met, the function returns early without attempting to modify the data.

This function does not increment [SlidingTilesData::slides](#). The caller is recommended to increment it based on if this function is true.

Definition at line 41 of file [SlidingTilesFunctions.cpp](#).

6.3.2.5 startGame()

```
void SlidingTilesFunctions::startGame (
    SlidingTilesEnums::Difficulty difficulty)
```

Reinitializes the data in [SlidingTilesData](#) to a reset state based on the `difficulty`, allowing for a new game of Sliding Tiles to be played.

Parameters

in	<i>difficulty</i>	The difficulty of the Sliding Tiles game being started.
----	-------------------	---

Author

Rajdeep Chowdhury

Date

8.04.2026

See also

[SlidingTilesData](#)

Definition at line 31 of file [SlidingTilesFunctions.cpp](#).

6.3.2.6 writeScore()

```
void SlidingTilesFunctions::writeScore (
    SlidingTilesEnums::Difficulty difficulty)
```

Appends the current number of [SlidingTilesData::slides](#) to the end of a text file representing the leaderboard of the given `difficulty`.

Parameters

in	<i>difficulty</i>	The difficulty of the Sliding Tiles game being played.
----	-------------------	--

Author

Raj Bahadur Bhat

Date

8.04.2026

Note

The amount of scores that can be in a leaderboard text file has no limit.

Definition at line 129 of file [SlidingTilesFunctions.cpp](#).

6.4 SlidingTilesFunctionsHelpers Namespace Reference

Namespace containing helper functions and a constant expression used by [SlidingTilesFunctions](#).

Functions

- [SlidingTilesEnums::Direction getRandomDirection](#) ()
Generates a random direction.
- [SlidingTilesEnums::Direction getOpposite](#) ([SlidingTilesEnums::Direction](#) dir)
Converts the direction given into its opposite direction.

Variables

- constexpr unsigned short [LEADERBOARD_SIZE](#) = 10
Constant representing the size of the leaderboard.

6.4.1 Detailed Description

Namespace of helper functions and a `constexpr` integer that the functions of [SlidingTilesFunctions](#) use.

Author

Rajdeep Chowdhury

Date

8.04.2026

See also

[SlidingTilesFunctions](#)

6.4.2 Function Documentation

6.4.2.1 getOpposite()

```
SlidingTilesEnums::Direction SlidingTilesFunctionsHelpers::getOpposite (
    SlidingTilesEnums::Direction dir)
```

Converts `dir` into the [SlidingTilesEnums::Direction](#) representation of the logical/common-sense opposite direction.

Parameters

in	<i>dir</i>	The direction which will be converted.
----	------------	--

Return values

<i>SlidingTilesEnums::Direction::UP</i>	If the direction given is <i>SlidingTilesEnums::Direction::DOWN</i>
<i>SlidingTilesEnums::Direction::DOWN</i>	If the direction given is <i>SlidingTilesEnums::Direction::UP</i>
<i>SlidingTilesEnums::Direction::LEFT</i>	If the direction given is <i>SlidingTilesEnums::Direction::RIGHT</i>
<i>SlidingTilesEnums::Direction::RIGHT</i>	If the direction given is <i>SlidingTilesEnums::Direction::LEFT</i>

Author

Fabio Bustamante Romillo

Date

8.04.2026

Definition at line 26 of file [SlidingTilesFunctionsHelpers.cpp](#).

6.4.2.2 getRandomDirection()

[*SlidingTilesEnums::Direction*](#) SlidingTilesFunctionsHelpers::getRandomDirection ()

Generates a random [*SlidingTilesEnums::Direction*](#)

Return values

<i>SlidingTilesEnums::Direction::UP</i>	1/4 chance.
<i>SlidingTilesEnums::Direction::DOWN</i>	1/4 chance.
<i>SlidingTilesEnums::Direction::LEFT</i>	1/4 chance.
<i>SlidingTilesEnums::Direction::RIGHT</i>	1/4 chance.

Author

Fabio Bustamante Romillo

Date

8.04.2026

Definition at line 21 of file [SlidingTilesFunctionsHelpers.cpp](#).

6.4.3 Variable Documentation**6.4.3.1 LEADERBOARD_SIZE**

unsigned short SlidingTilesFunctionsHelpers::LEADERBOARD_SIZE = 10 [constexpr]

Constant integer representing how many scores are loaded by the leaderboard.

Definition at line 32 of file [SlidingTilesFunctionsHelpers.hpp](#).

6.5 UserFunctions Namespace Reference

Namespace containing functions that the user interacts with.

Functions

- bool [selectDifficulty](#) ()
Allows the user to select their Sliding Tiles game difficulty.
- bool [performSlide](#) ()
Allows the user to perform a slide in a Sliding Tiles game.

6.5.1 Detailed Description

Namespace containing functions that turn user input into output. It is the module that users directly interact with.

Author

Rajdeep Chowdhury

Date

8.04.2026

6.5.2 Function Documentation

6.5.2.1 [performSlide\(\)](#)

```
bool UserFunctions::performSlide ()
```

Prompts the user to slide the empty tile in all [SlidingTilesEnums::Direction](#) available or to quit the game they are currently in. Sliding will affect the game state and end the game if the win condition is met.

Return values

<i>true</i>	If quit is not selected or the game is not over.
<i>false</i>	If quit is selected
<i>false</i>	If the game determines it can end.

Author

Rajdeep Chowdhury

Date

8.04.2026

Definition at line 67 of file [UserFunctions.cpp](#).

6.5.2.2 selectDifficulty()

```
bool UserFunctions::selectDifficulty ()
```

Prompts the user to select their [SlidingTilesEnums::Difficulty](#) or to quit. Selecting a difficulty will begin a game afterwards. Quitting will end program execution.

Return values

<i>true</i>	If quit is not selected.
<i>false</i>	If quit is selected.

Author

Rajdeep Chowdhury

Date

8.04.2026

Definition at line 29 of file [UserFunctions.cpp](#).

6.6 UserFunctionsHelpers Namespace Reference

Namespace containing helper functions used by [UserFunctions](#).

Functions

- void [startGame](#) ()
Initializes data to start a game of Sliding Tiles.
- void [trySlide](#) ([SlidingTilesEnums::Direction](#) direction)
Tries to slide the empty tile in the given direction.
- void [printBoard](#) ()
Print the Sliding Tiles board.
- void [printLeaderboard](#) ()
Print the Sliding Tiles leaderboard for the current difficulty.
- bool [continueGame](#) ()
Determines whether the Sliding Tiles game should continue.

6.6.1 Detailed Description

Namespace containing helper functions that the functions of [UserFunctions](#) use.

Author

Rajdeep Chowdhury

Date

8.04.2026

See also

[UserFunctions](#)

6.6.2 Function Documentation

6.6.2.1 continueGame()

```
bool UserFunctionsHelpers::continueGame ()
```

Determines whether the Sliding Tiles game should continue based on if the board is ordered or not. If the board is ordered, it will also handle applying the effect of winning the game.

Return values

<i>true</i>	If the game must still continue.
<i>false</i>	If the game should end.

Author

Rajdeep Chowdhury

Date

8.04.2026

Definition at line 62 of file [UserFunctionsHelpers.cpp](#).

6.6.2.2 printBoard()

```
void UserFunctionsHelpers::printBoard ()
```

Displays all the tiles of the Sliding Tiles board onto the console.

Author

Rajdeep Chowdhury

Date

8.04.2026

Definition at line 39 of file [UserFunctionsHelpers.cpp](#).

6.6.2.3 printLeaderboard()

```
void UserFunctionsHelpers::printLeaderboard ()
```

Displays the top [SlidingTilesFunctionsHelpers::LEADERBOARD_SIZE](#) scores achieved for the [SlidingTilesData::currentDifficulty](#) being played.

Author

Rajdeep Chowdhury

Date

8.04.2026

See also

[SlidingTilesFunctionsHelpers](#)

[SlidingTilesData](#)

Definition at line 52 of file [UserFunctionsHelpers.cpp](#).

6.6.2.4 startGame()

```
void UserFunctionsHelpers::startGame ()
```

Uses functions from [SlidingTilesFunctions](#) to initialize the data required to start a game of Sliding Tiles.

Author

Rajdeep Chowdhury

Date

8.04.2026

Definition at line 26 of file [UserFunctionsHelpers.cpp](#).

6.6.2.5 trySlide()

```
void UserFunctionsHelpers::trySlide (
    SlidingTilesEnums::Direction direction)
```

Tries to slide the empty tile in the given directionm handling both the success and failure cases.

Parameters

in	<i>direction</i>	The direction the slide will occur.
----	------------------	-------------------------------------

Author

Rajdeep Chowdhury

Date

8.04.2026

Definition at line 32 of file [UserFunctionsHelpers.cpp](#).

Chapter 7

Class Documentation

7.1 Board Class Reference

A class that represents a board.

```
#include <Board.hpp>
```

Public Member Functions

- [Board](#) () noexcept
Board object constructor.
- [Board](#) ([SlidingTilesEnums::Difficulty](#) difficulty) noexcept
Board object constructor.
- bool [hasEmptyRow](#) () const noexcept
Checks if the board has an empty row or is empty.
- bool [canAccess](#) (size_t row, size_t column) const noexcept
Checks if a Board index can be accessed. Index starts at 0.
- bool [canAccess](#) (size_t row) const noexcept
Checks if a Board row can be accessed. Index starts at 0.
- size_t & [access](#) (size_t row, size_t column)
Accesses an integer in the board. Indexing starts at 0.
- std::vector< size_t > & [access](#) (size_t row)
Accesses a row in the board. Indexing starts at 0.

Protected Attributes

- std::vector< std::vector< size_t > > [board](#) {}
The data representation of the board.

7.1.1 Detailed Description

A class that represents a board for any game that can be represented by numbered tiles.

Author

Rajdeep Chowdhury

Date

8.04.2026

Definition at line 26 of file [Board.hpp](#).

7.1.2 Constructor & Destructor Documentation

7.1.2.1 Board() [1/2]

```
Board::Board () [noexcept]
```

The default constructor for the [Board](#) that creates an empty, sizeless board.

Author

Rajdeep Chowdhury

Date

8.04.2026

Definition at line 18 of file [Board.cpp](#).

7.1.2.2 Board() [2/2]

```
Board::Board (  
    SlidingTilesEnums::Difficulty difficulty) [noexcept]
```

The parameterized constructor for the [Board](#) that creates a board based on the difficulty passed through for Sliding Tiles.

Parameters

in	<i>difficulty</i>	The difficulty of the Sliding Tiles game desired.
----	-------------------	---

Author

Raj Bahadur Bhat

Date

8.04.2026

See also

[SlidingTilesEnums](#)

Definition at line 20 of file [Board.cpp](#).

7.1.3 Member Function Documentation

7.1.3.1 access() [1/2]

```
std::vector< size_t > & Board::access (
    size_t row)
```

Returns a reference to the `size_t` row vector at the specified `row` index.

Parameters

<code>in</code>	<code>row</code>	The row of the board being accessed
-----------------	------------------	-------------------------------------

Returns

A reference to the row vector on the board.

Exceptions

<code>std::out_of_range</code>	If the row being accessed does not exist.
--------------------------------	---

Author

Rajdeep Chowdhury

Date

8.04.2026

Note

The user should use the [Board::canAccess\(size_t\) const](#) method before using this method to avoid the exception being thrown.

See also

[Board::canAccess\(size_t\) const](#)

Definition at line 44 of file [Board.cpp](#).

7.1.3.2 access() [2/2]

```
size_t & Board::access (  
    size_t row,  
    size_t column)
```

Returns a reference to the `size_t` integer at the specified `row` and `column` index.

Parameters

in	<i>row</i>	The row of the board being accessed
in	<i>column</i>	The column of the board being accessed.

Returns

A reference to the `size_t` integer on the board.

Exceptions

<i>std::out_of_range</i>	If the row or column being accessed does not exist.
--------------------------	---

Author

Rajdeep Chowdhury

Date

8.04.2026

Note

The user should use the [Board::canAccess\(size_t, size_t\) const](#) method before using this method to avoid the exception being thrown.

See also

[Board::canAccess\(size_t, size_t\) const](#)

Definition at line 48 of file [Board.cpp](#).

7.1.3.3 canAccess() [1/2]

```
bool Board::canAccess (
    size_t row) const [noexcept]
```

Checks if a [Board](#) row can exist in accordance to the size of the vector representing the board rows. Index starts at 0. This method is made to compliment using [Board::access\(size_t\)](#) to prevent an exception being thrown.

Parameters

in	<i>row</i>	The row of the board being accessed.
----	------------	--------------------------------------

Return values

<i>true</i>	If the row exists.
<i>false</i>	If the row does not exist.

Author

Rajdeep Chowdhury

Date

8.04.2026

See also

[Board::access\(size_t\)](#)

Definition at line 52 of file [Board.cpp](#).

7.1.3.4 canAccess() [2/2]

```
bool Board::canAccess (
    size_t row,
    size_t column) const [noexcept]
```

Checks if a [Board](#) index can exist in accordance to the size of the vectors representing the board. Index starts at 0. This method is made to compliment using [Board::access\(size_t, size_t\)](#) to prevent an exception being thrown.

Parameters

in	<i>row</i>	The row of the board being accessed.
in	<i>column</i>	The coloum of the board being accessed.

Return values

<i>true</i>	If the index exists.
<i>false</i>	If the index does not exist.

Author

Rajdeep Chowdhury

Date

8.04.2026

See also

[Board::access\(size_t\)](#)

Definition at line 56 of file [Board.cpp](#).

7.1.3.5 hasEmptyRow()

```
bool Board::hasEmptyRow () const [noexcept]
```

Checks if one of the vector sizes of a board row is 0. Also checks for if the board itself has no rows.

Return values

<i>true</i>	If an empty row exists.
<i>true</i>	If the board itself is empty.
<i>false</i>	If neither condition is true.

Author

Rajdeep Chowdhury

Date

8.04.2026

Definition at line 36 of file [Board.cpp](#).

7.1.4 Member Data Documentation**7.1.4.1 board**

```
std::vector<std::vector<size_t> > Board::board {} [protected]
```

The vector of vectors of size_t which represents a board.

Definition at line 30 of file [Board.hpp](#).

The documentation for this class was generated from the following files:

- [Board.hpp](#)
- [Board.cpp](#)

Chapter 8

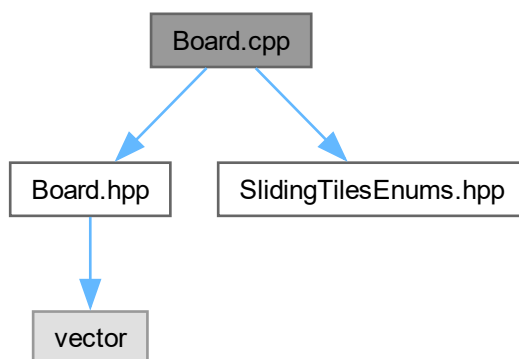
File Documentation

8.1 Board.cpp File Reference

File containing the initialization of all [Board](#) methods.

```
#include "Board.hpp"  
#include "SlidingTilesEnums.hpp"
```

Include dependency graph for Board.cpp:



8.1.1 Detailed Description

This file contains the initialization of all [Board](#) methods declared in [Board.hpp](#).

Author

Rajdeep Chowdhury

Raj Bahadur Bhat

Date

8.04.2026

See also[Board.hpp](#)[SlidingTilesEnums.hpp](#)Definition in file [Board.cpp](#).

8.2 Board.cpp

[Go to the documentation of this file.](#)

```

00001
00015 #include "Board.hpp"
00016 #include "SlidingTilesEnums.hpp"
00017
00018 Board::Board() noexcept {}
00019
00020 Board::Board(SlidingTilesEnums::Difficulty difficulty) noexcept {
00021     int size = static_cast<int>(difficulty); //Convert the enum into the board size
00022
00023     //Build the board
00024     board.resize(size);
00025     int counter = 1;
00026     for (int col = 0; col < size; col++) {
00027         board.at(col).resize(size);
00028
00029         for (int row = 0; row < size; row++) {
00030             board.at(col).at(row) = counter;
00031             counter++;
00032         }
00033     }
00034 }
00035
00036 bool Board::hasEmptyRow() const noexcept {
00037     if (this->board.size() <= 0) return true; //Check if board itself is empty
00038     for (int i = 0; i < board.size(); i++) { //Check each row for an empty row
00039         if (board.at(i).size() <= 0) return true;
00040     }
00041     return false;
00042 }
00043
00044 std::vector<size_t>& Board::access(size_t row) {
00045     return this->board.at(row);
00046 }
00047
00048 size_t& Board::access(size_t row, size_t column) {
00049     return this->access(row).at(column);
00050 }
00051
00052 bool Board::canAccess(size_t row) const noexcept {
00053     return (row >= 0 && row < this->board.size());
00054 }
00055
00056 bool Board::canAccess(size_t row, size_t column) const noexcept {
00057     if (!canAccess(row)) return false;
00058     size_t size = this->board.at(row).size();
00059     return (column >= 0 && column < size);
00060 }

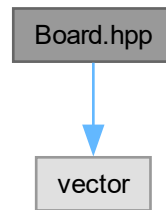
```

8.3 Board.hpp File Reference

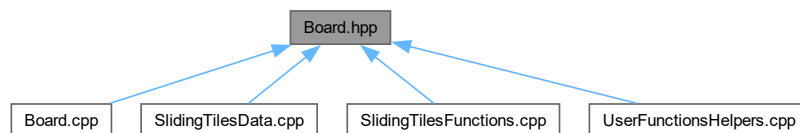
File containing a [Board](#) class.

```
#include <vector>
```

Include dependency graph for Board.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [Board](#)
A class that represents a board.

Namespaces

- namespace [SlidingTilesEnums](#)
Namespace containing enum classes for a Sliding Tiles game.

8.3.1 Detailed Description

This file contains a class representing a board for any game, with the declaration of its method and attributes initialized.

Author

Rajdeep Chowdhury

Date

8.04.2026

See also

[Board.cpp](#)

Definition in file [Board.hpp](#).

8.4 Board.hpp

[Go to the documentation of this file.](#)

```

00001
00013 #pragma once
00014 #include <vector>
00015
00016 //Forward declaration
00017
00018 namespace SlidingTilesEnums {
00019     enum class Difficulty;
00020 }
00021
00026 class Board {
00027 protected:
00030     std::vector<std::vector<size_t>> board{};
00031
00032 public:
00037     Board() noexcept;
00038
00047     Board(SlidingTilesEnums::Difficulty difficulty) noexcept;
00048
00057     bool hasEmptyRow() const noexcept;
00058
00071     bool canAccess(size_t row, size_t column) const noexcept;
00072
00084     bool canAccess(size_t row) const noexcept;
00085
00099     size_t& access(size_t row, size_t column);
00100
00113     std::vector<size_t>& access(size_t row);
00114 };

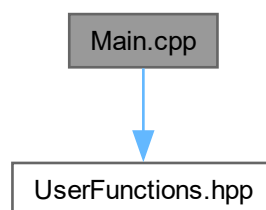
```

8.5 Main.cpp File Reference

Main file where the program is executed.

```
#include "UserFunctions.hpp"
```

Include dependency graph for Main.cpp:



Functions

- int `main` ()

Execute the program.

8.5.1 Detailed Description

This file is the main file. It is where the functions that start the program are called.

Author

Rajdeep Chowdhury

Date

8.04.2026

See also

[UserFunctions.hpp](#)

Definition in file [Main.cpp](#).

8.5.2 Function Documentation

8.5.2.1 main()

```
int main ()
```

The main function that executes the program.

Returns

Exit code.

Author

Rajdeep Chowdhury

Date

8.04.2026

Definition at line 19 of file [Main.cpp](#).

8.6 Main.cpp

[Go to the documentation of this file.](#)

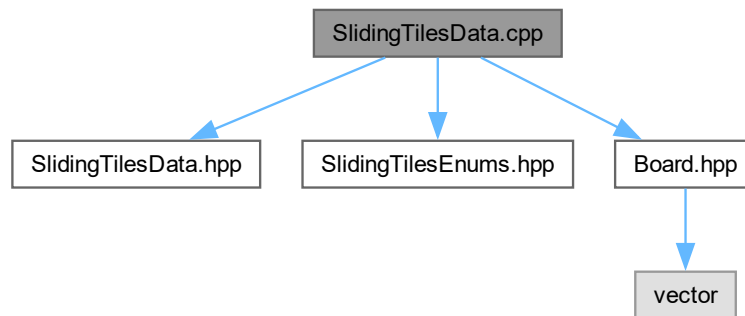
```
00001
00012 #include "UserFunctions.hpp"
00013
00019 int main() {
00020     while (UserFunctions::selectDifficulty()) {
00021         while (UserFunctions::performSlide());
00022     }
00023 }
```

8.7 SlidingTilesData.cpp File Reference

File containing sliding tiles variable initializations.

```
#include "SlidingTilesData.hpp"
#include "SlidingTilesEnums.hpp"
#include "Board.hpp"
```

Include dependency graph for SlidingTilesData.cpp:



Namespaces

- namespace [SlidingTilesData](#)
Namespace containing data for a Sliding Tiles game.

Variables

- [Board](#) [SlidingTilesData::board](#)
The [Board](#) object for Sliding Tiles.
- unsigned int [SlidingTilesData::slides](#) = 0
The amount of slides performed.
- size_t [SlidingTilesData::currentRow](#) = 0
The current row index of the empty tile.
- size_t [SlidingTilesData::currentColumn](#) = 0
The current column index of the empty tile.
- size_t [SlidingTilesData::boardSize](#) = 0
The size of the board in a single direction.
- [SlidingTilesEnums::Difficulty](#) [SlidingTilesData::currentDifficulty](#) = [SlidingTilesEnums::Difficulty::EASY](#)
The current difficulty being played.

8.7.1 Detailed Description

This file contains the initializations of the externally linked [SlidingTilesData](#) namespace variables found in [SlidingTilesData.hpp](#).

Author

Rajdeep Chowdhury

Date

8.04.2026

See also

[SlidingTilesData.hpp](#)

[SlidingTilesEnums.hpp](#)

[Board.hpp](#)

Definition in file [SlidingTilesData.cpp](#).

8.8 SlidingTilesData.cpp

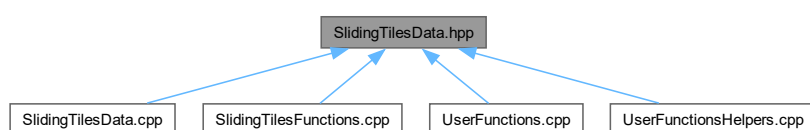
[Go to the documentation of this file.](#)

```
00001
00015 #include "SlidingTilesData.hpp"
00016 #include "SlidingTilesEnums.hpp"
00017 #include "Board.hpp"
00018
00019 namespace SlidingTilesData {
00020     Board board;
00021     unsigned int slides = 0;
00022     size_t currentRow = 0, currentColumn = 0;
00023     size_t boardSize = 0;
00024     SlidingTilesEnums::Difficulty currentDifficulty = SlidingTilesEnums::Difficulty::EASY;
00025 }
```

8.9 SlidingTilesData.hpp File Reference

File containing a namespace data for a Sliding Tiles game.

This graph shows which files directly or indirectly include this file:



Namespaces

- namespace [SlidingTilesEnums](#)
Namespace containing enum classes for a Sliding Tiles game.
- namespace [SlidingTilesData](#)
Namespace containing data for a Sliding Tiles game.

8.9.1 Detailed Description

This file contains a namespace with external links to data that change and persist during a Sliding Tiles game.

Author

Rajdeep Chowdhury

Date

8.04.2026

See also

[SlidingTilesData.cpp](#)

Definition in file [SlidingTilesData.hpp](#).

8.10 SlidingTilesData.hpp

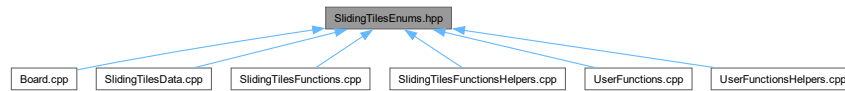
[Go to the documentation of this file.](#)

```
00001
00013 #pragma once
00014
00015 //Forward declarations
00016
00017 class Board;
00018
00019 namespace SlidingTilesEnums {
00020     enum class Difficulty;
00021 }
00022
00028 namespace SlidingTilesData {
00031     extern Board board;
00032
00035     extern unsigned int slides;
00036
00039     extern size_t currentRow;
00040
00043     extern size_t currentColumn;
00044
00047     extern size_t boardSize;
00048
00051     extern SlidingTilesEnums::Difficulty currentDifficulty;
00052 }
00053
```

8.11 SlidingTilesEnums.hpp File Reference

File containing a namespace with enum classes for a Sliding Tiles game.

This graph shows which files directly or indirectly include this file:



Namespaces

- namespace [SlidingTilesEnums](#)
Namespace containing enum classes for a Sliding Tiles game.

Enumerations

- enum class [SlidingTilesEnums::Difficulty](#) { [SlidingTilesEnums::EASY](#) = 3 , [SlidingTilesEnums::MEDIUM](#) = 4 , [SlidingTilesEnums::HARD](#) = 6 , [SlidingTilesEnums::INSANE](#) = 9 }
Enum class containing difficulties for a Sliding Tiles game.
- enum class [SlidingTilesEnums::Direction](#) { [SlidingTilesEnums::UP](#) , [SlidingTilesEnums::DOWN](#) , [SlidingTilesEnums::LEFT](#) , [SlidingTilesEnums::RIGHT](#) }
Enum class containing directions that a slide can occur.

8.11.1 Detailed Description

This file contains a namespace with enum classes that represent the difficulty of a Sliding Tiles game and the directions which a slide can occur.

Author

Rajdeep Chowdhury

Date

8.04.2026

Definition in file [SlidingTilesEnums.hpp](#).

8.12 SlidingTilesEnums.hpp

[Go to the documentation of this file.](#)

```

00001
00011 #pragma once
00012
00018 namespace SlidingTilesEnums {
00024     enum class Difficulty {
00028         EASY = 3,
00029
00033         MEDIUM = 4,
00034
00038         HARD = 6,
00039
00043         INSANE = 9
00044     };
00045
00051     enum class Direction {
00054         UP,
00055
00058         DOWN,
00059
00062         LEFT,
00063
00066         RIGHT
00067     };
00068 }

```

8.13 SlidingTilesFunctions.cpp File Reference

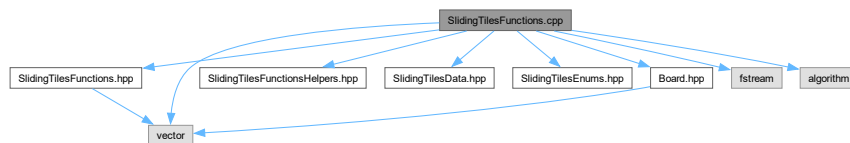
File containing the initialization of the [SlidingTilesFunctions](#) functions.

```

#include "SlidingTilesFunctions.hpp"
#include "SlidingTilesFunctionsHelpers.hpp"
#include "SlidingTilesData.hpp"
#include "SlidingTilesEnums.hpp"
#include "Board.hpp"
#include <vector>
#include <fstream>
#include <algorithm>

```

Include dependency graph for SlidingTilesFunctions.cpp:



Namespaces

- namespace [SlidingTilesFunctions](#)

Namespace containing the functions used during a game of Sliding Tiles.

Functions

- void [SlidingTilesFunctions::startGame](#) ([SlidingTilesEnums::Difficulty](#) difficulty)
Sets up the data required to start a Sliding Tiles game.
- bool [SlidingTilesFunctions::slide](#) ([SlidingTilesEnums::Direction](#) direction)
Attempts to slide the empty tile to an adjacent filled tile.
- void [SlidingTilesFunctions::shuffle](#) ()
Shuffles the board.
- bool [SlidingTilesFunctions::isBoardOrdered](#) ()
Checks if all the board's numbers are ordered.
- void [SlidingTilesFunctions::writeScore](#) ([SlidingTilesEnums::Difficulty](#) difficulty)
Writes the number of slides to its associated leaderboard.
- std::vector< unsigned int > [SlidingTilesFunctions::loadScores](#) ([SlidingTilesEnums::Difficulty](#) difficulty)
Loads the top scores from its associated leaderboard.

8.13.1 Detailed Description

This file contains the initialization of the [SlidingTilesFunctions](#) functions declared in [SlidingTilesFunctions.hpp](#).

Author

Rajdeep Chowdhury
Fabio Bustamante Romillo
Raj Bahadur Bhat

Date

9.04.2026

See also

[SlidingTilesFunctions.hpp](#)
[SlidingTilesFunctionsHelpers.hpp](#)
[SlidingTilesData.hpp](#)
[SlidingTilesEnums.hpp](#)
[Board.hpp](#)
[SlidingTilesData](#)

Definition in file [SlidingTilesFunctions.cpp](#).

8.14 SlidingTilesFunctions.cpp

[Go to the documentation of this file.](#)

```

00001
00020 #include "SlidingTilesFunctions.hpp"
00021 #include "SlidingTilesFunctionsHelpers.hpp"
00022 #include "SlidingTilesData.hpp"
00023 #include "SlidingTilesEnums.hpp"
00024 #include "Board.hpp"
00025 #include <vector>
00026 #include <fstream>
00027 #include <algorithm>
00028
00029 namespace SlidingTilesFunctions {
00030
00031     void startGame(SlidingTilesEnums::Difficulty difficulty) {
00032         SlidingTilesData::board = Board(difficulty);
00033         SlidingTilesData::boardSize = static_cast<int>(difficulty);
00034         SlidingTilesData::currentColumn =
00035             SlidingTilesData::currentRow =
00036             SlidingTilesData::boardSize - 1;
00037         SlidingTilesData::slides = 0;
00038         SlidingTilesData::currentDifficulty = difficulty;
00039     }
00040
00041     bool slide(SlidingTilesEnums::Direction direction) {
00042         using namespace SlidingTilesData;
00043         bool canSlide = false;
00044         //The main purpose of the below check is to ensure the board is not empty, meaning the default
board is the current board.
00045         if (board.hasEmptyRow()) return false;
00046
00047         //Determine if a slide can occur in the desired direction
00048         switch (direction) {
00049             case SlidingTilesEnums::Direction::DOWN:
00050                 canSlide = board.canAccess(currentRow + 1, currentColumn);
00051                 break;
00052             case SlidingTilesEnums::Direction::UP:
00053                 canSlide = board.canAccess(currentRow - 1, currentColumn);
00054                 break;
00055             case SlidingTilesEnums::Direction::LEFT:
00056                 canSlide = board.canAccess(currentRow, currentColumn - 1);
00057                 break;
00058             case SlidingTilesEnums::Direction::RIGHT:
00059                 canSlide = board.canAccess(currentRow, currentColumn + 1);
00060                 break;
00061         }
00062
00063         if (!canSlide) return false;
00064
00065         //Slide the desired square to the empty square
00066         switch (direction) {
00067             case SlidingTilesEnums::Direction::DOWN:
00068                 std::swap(board.access(currentRow, currentColumn), board.access(currentRow + 1,
currentColumn));
00069                 currentRow += 1;
00070                 break;
00071             case SlidingTilesEnums::Direction::UP:
00072                 std::swap(board.access(currentRow, currentColumn), board.access(currentRow - 1,
currentColumn));
00073                 currentRow -= 1;
00074                 break;
00075             case SlidingTilesEnums::Direction::LEFT:
00076                 std::swap(board.access(currentRow, currentColumn), board.access(currentRow, currentColumn
- 1));
00077                 currentColumn -= 1;
00078                 break;
00079             case SlidingTilesEnums::Direction::RIGHT:
00080                 std::swap(board.access(currentRow, currentColumn), board.access(currentRow, currentColumn
+ 1));
00081                 currentColumn += 1;
00082                 break;
00083         }
00084         return true;
00085     }
00086
00087     void shuffle() {
00088         std::srand(static_cast<unsigned int>(std::time(nullptr)));
00089
00090         size_t moves = SlidingTilesData::boardSize * SlidingTilesData::boardSize * 10;
00091
00092         SlidingTilesEnums::Direction lastMove = SlidingTilesEnums::Direction::UP;
00093         bool hasLastMove = false;
00094         int i = 0;
00095         while (i < moves) {

```



```

00096         SlidingTilesEnums::Direction move = SlidingTilesFunctionsHelpers::getRandomDirection();
00097
00098         // validMove = true only if move is not opposite and slide succeeds
00099         bool validMove = (!hasLastMove || move !=
SlidingTilesFunctionsHelpers::getOpposite(lastMove)) && slide(move);
00100
00101         // increment counter if valid
00102         i += validMove; // true = 1, false = 0
00103         lastMove = validMove ? move : lastMove; // update lastMove only if valid
00104         hasLastMove = hasLastMove || validMove; // mark hasLastMove if any move succeeded
00105     }
00106
00107     // Move empty tile to bottom right
00108     while (SlidingTilesData::currentRow < SlidingTilesData::boardSize - 1)
00109         slide(SlidingTilesEnums::Direction::DOWN);
00110     while (SlidingTilesData::currentColumn < SlidingTilesData::boardSize - 1)
00111         slide(SlidingTilesEnums::Direction::RIGHT);
00112 }
00113
00114 bool isBoardOrdered() {
00115     bool isOrdered = true;
00116
00117     for (size_t row = 0; row < SlidingTilesData::boardSize; row++) {
00118         for (size_t col = 0; col < SlidingTilesData::boardSize; col++) {
00119
00120             size_t expectedValue = row * SlidingTilesData::boardSize + col + 1;
00121
00122             isOrdered = isOrdered && (SlidingTilesData::board.access(row, col) == expectedValue);
00123         }
00124     }
00125
00126     return isOrdered;
00127 }
00128
00129 void writeScore(SlidingTilesEnums::Difficulty difficulty) {
00130     // Get filename based on difficulty
00131     std::string filename;
00132     switch (difficulty) {
00133     case SlidingTilesEnums::Difficulty::EASY:
00134         filename = "scores_easy.txt";
00135         break;
00136     case SlidingTilesEnums::Difficulty::MEDIUM:
00137         filename = "scores_medium.txt";
00138         break;
00139     case SlidingTilesEnums::Difficulty::HARD:
00140         filename = "scores_hard.txt";
00141         break;
00142     case SlidingTilesEnums::Difficulty::INSANE:
00143         filename = "scores_insane.txt";
00144         break;
00145     }
00146
00147     // Open file to append score
00148     std::ofstream outFile(filename, std::ios::app);
00149     if (outFile.is_open()) {
00150         outFile << SlidingTilesData::slides << std::endl;
00151         outFile.close();
00152     }
00153 }
00154
00155 std::vector<unsigned int> loadScores(SlidingTilesEnums::Difficulty difficulty) {
00156     std::vector<unsigned int> scores;
00157
00158     // Get filename based on difficulty
00159     std::string filename;
00160     switch (difficulty) {
00161     case SlidingTilesEnums::Difficulty::EASY:
00162         filename = "scores_easy.txt";
00163         break;
00164     case SlidingTilesEnums::Difficulty::MEDIUM:
00165         filename = "scores_medium.txt";
00166         break;
00167     case SlidingTilesEnums::Difficulty::HARD:
00168         filename = "scores_hard.txt";
00169         break;
00170     case SlidingTilesEnums::Difficulty::INSANE:
00171         filename = "scores_insane.txt";
00172         break;
00173     }
00174
00175     // Read all scores from file
00176     std::ifstream inFile(filename);
00177     if (inFile.is_open()) {
00178         unsigned int score;
00179         while (inFile >> score) {
00180             scores.push_back(score);
00181         }
00182     }

```

```

00182         inFile.close();
00183     }
00184
00185     // Sort scores (lowest is best)
00186     std::sort(scores.begin(), scores.end());
00187
00188     // Keep only top 10
00189     if (scores.size() > SlidingTilesFunctionsHelpers::LEADERBOARD_SIZE) {
00190         scores.resize(SlidingTilesFunctionsHelpers::LEADERBOARD_SIZE);
00191     }
00192
00193     return scores;
00194 }
00195 }

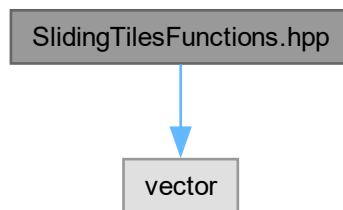
```

8.15 SlidingTilesFunctions.hpp File Reference

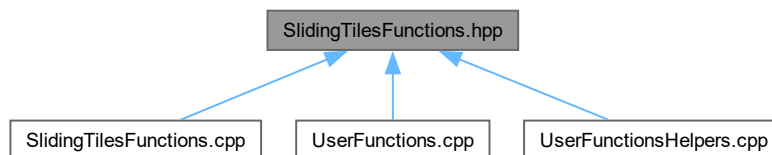
File containing a namespace with declared functions used during a game of Sliding Tiles.

```
#include <vector>
```

Include dependency graph for SlidingTilesFunctions.hpp:



This graph shows which files directly or indirectly include this file:



Namespaces

- namespace [SlidingTilesEnums](#)
Namespace containing enum classes for a Sliding Tiles game.
- namespace [SlidingTilesFunctions](#)
Namespace containing the functions used during a game of Sliding Tiles.

Functions

- void [SlidingTilesFunctions::startGame](#) ([SlidingTilesEnums::Difficulty](#) difficulty)
Sets up the data required to start a Sliding Tiles game.
- bool [SlidingTilesFunctions::slide](#) ([SlidingTilesEnums::Direction](#) direction)
Attempts to slide the empty tile to an adjacent filled tile.
- void [SlidingTilesFunctions::shuffle](#) ()
Shuffles the board.
- bool [SlidingTilesFunctions::isBoardOrdered](#) ()
Checks if all the board's numbers are ordered.
- void [SlidingTilesFunctions::writeScore](#) ([SlidingTilesEnums::Difficulty](#) difficulty)
Writes the number of slides to its associated leaderboard.
- std::vector< unsigned int > [SlidingTilesFunctions::loadScores](#) ([SlidingTilesEnums::Difficulty](#) difficulty)
Loads the top scores from its associated leaderboard.

8.15.1 Detailed Description

This file contains a namespace with the declaration of functions which are called during a game of Sliding Tiles to manipulate the data of [SlidingTilesData](#).

Author

Rajdeep Chowdhury

Date

8.04.2026

See also

[SlidingTilesFunctions.cpp](#)

[SlidingTilesData](#)

Definition in file [SlidingTilesFunctions.hpp](#).

8.16 SlidingTilesFunctions.hpp

[Go to the documentation of this file.](#)

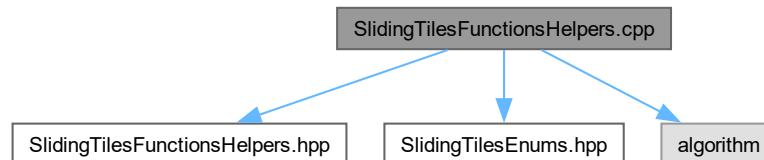
```
00001
00014 #pragma once
00015 #include <vector>
00016
00017 //Forward declarations
00018
00019 namespace SlidingTilesEnums {
00020     enum class Difficulty;
00021     enum class Direction;
00022 }
00023
00031 namespace SlidingTilesFunctions {
00040     void startGame(SlidingTilesEnums::Difficulty difficulty);
00041
00058     bool slide(SlidingTilesEnums::Direction direction);
00059
00078     void shuffle();
00079
00087     bool isBoardOrdered();
00088
00097     void writeScore(SlidingTilesEnums::Difficulty difficulty);
00098
00109     std::vector<unsigned int> loadScores(SlidingTilesEnums::Difficulty difficulty);
00110 }
```

8.17 SlidingTilesFunctionsHelpers.cpp File Reference

File containing the initialization of the [SlidingTilesFunctionsHelpers](#) functions.

```
#include "SlidingTilesFunctionsHelpers.hpp"
#include "SlidingTilesEnums.hpp"
#include <algorithm>
```

Include dependency graph for SlidingTilesFunctionsHelpers.cpp:



Namespaces

- namespace [SlidingTilesFunctionsHelpers](#)
Namespace containing helper functions and a constant expression used by [SlidingTilesFunctions](#).

Functions

- [SlidingTilesEnums::Direction SlidingTilesFunctionsHelpers::getRandomDirection \(\)](#)
Generates a random direction.
- [SlidingTilesEnums::Direction SlidingTilesFunctionsHelpers::getOpposite \(SlidingTilesEnums::Direction dir\)](#)
Converts the direction given into its opposite direction.

8.17.1 Detailed Description

This file contains the initialization of the [SlidingTilesFunctionsHelpers](#) helper functions declared in [SlidingTilesFunctionsHelpers.hpp](#).

Author

Rajdeep Chowdhury
 Fabio Bustamante Romillo

Date

8.04.2026

See also

[SlidingTilesFunctionsHelpers.hpp](#)
[SlidingTilesEnums.hpp](#)
[SlidingTilesFunctions](#)

Definition in file [SlidingTilesFunctionsHelpers.cpp](#).

8.18 SlidingTilesFunctionsHelpers.cpp

[Go to the documentation of this file.](#)

```

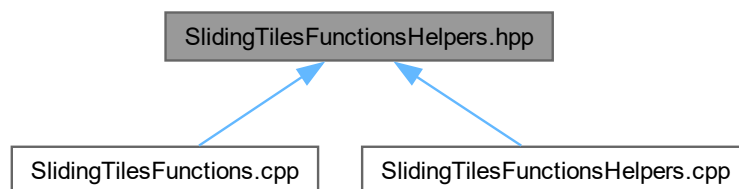
00001
00016 #include "SlidingTilesFunctionsHelpers.hpp"
00017 #include "SlidingTilesEnums.hpp"
00018 #include <algorithm>
00019
00020 namespace SlidingTilesFunctionsHelpers {
00021     SlidingTilesEnums::Direction getRandomDirection() {
00022         int r = std::rand() % 4;
00023         return static_cast<SlidingTilesEnums::Direction>(r);
00024     }
00025
00026     SlidingTilesEnums::Direction getOpposite(SlidingTilesEnums::Direction dir) {
00027         switch (dir) {
00028             case SlidingTilesEnums::Direction::UP:    return SlidingTilesEnums::Direction::DOWN;
00029             case SlidingTilesEnums::Direction::DOWN:  return SlidingTilesEnums::Direction::UP;
00030             case SlidingTilesEnums::Direction::LEFT:  return SlidingTilesEnums::Direction::RIGHT;
00031             case SlidingTilesEnums::Direction::RIGHT: return SlidingTilesEnums::Direction::LEFT;
00032         }
00033         return SlidingTilesEnums::Direction::UP; // fallback
00034     }
00035 }

```

8.19 SlidingTilesFunctionsHelpers.hpp File Reference

File containing a namespace of declared helper functions and a constant expression for [SlidingTilesFunctions](#).

This graph shows which files directly or indirectly include this file:



Namespaces

- namespace [SlidingTilesEnums](#)
Namespace containing enum classes for a Sliding Tiles game.
- namespace [SlidingTilesFunctionsHelpers](#)
Namespace containing helper functions and a constant expression used by [SlidingTilesFunctions](#).

Functions

- [SlidingTilesEnums::Direction SlidingTilesFunctionsHelpers::getRandomDirection \(\)](#)
Generates a random direction.
- [SlidingTilesEnums::Direction SlidingTilesFunctionsHelpers::getOpposite \(SlidingTilesEnums::Direction dir\)](#)
Converts the direction given into its opposite direction.

Variables

- constexpr unsigned short [SlidingTilesFunctionsHelpers::LEADERBOARD_SIZE](#) = 10
Constant representing the size of the leaderboard.

8.19.1 Detailed Description

This file contains a namespace of declared helper functions and a `constexpr` integer that the functions of [SlidingTilesFunctions](#) use.

Author

Rajdeep Chowdhury

Date

8.04.2026

See also

[SlidingTilesFunctionsHelpers.cpp](#)

[SlidingTilesFunctions](#)

Definition in file [SlidingTilesFunctionsHelpers.hpp](#).

8.20 SlidingTilesFunctionsHelpers.hpp

[Go to the documentation of this file.](#)

```
00001
00014 #pragma once
00015
00016 //Forward declaration
00017
00018 namespace SlidingTilesEnums {
00019     enum class Direction;
00020 }
00021
00029 namespace SlidingTilesFunctionsHelpers {
00032     constexpr unsigned short LEADERBOARD_SIZE = 10;
00033
00042     SlidingTilesEnums::Direction getRandomDirection();
00043
00054     SlidingTilesEnums::Direction getOpposite(SlidingTilesEnums::Direction dir);
00055 }
```

8.21 UserFunctions.cpp File Reference

File containing the initialization of the [UserFunctions](#) functions.

```
#include "UserFunctions.hpp"
#include "UserFunctionsHelpers.hpp"
#include "SlidingTilesFunctions.hpp"
#include "SlidingTilesData.hpp"
#include "SlidingTilesEnums.hpp"
#include <string>
#include <iostream>
#include <algorithm>
```

Include dependency graph for UserFunctions.cpp:



Namespaces

- namespace [UserFunctions](#)
Namespace containing functions that the user interacts with.

Functions

- bool [UserFunctions::selectDifficulty](#) ()
Allows the user to select their Sliding Tiles game difficulty.
- bool [UserFunctions::performSlide](#) ()
Allows the user to perform a slide in a Sliding Tiles game.

8.21.1 Detailed Description

This file contains the initialization of the [UserFunctions](#) functions declared in [UserFunctions.hpp](#).

Author

Rajdeep Chowdhury

Date

9.04.2026

See also

[UserFunctions.hpp](#)
[UserFunctionsHelpers.hpp](#)
[SlidingTilesFunctions.hpp](#)
[SlidingTilesData.hpp](#)
[SlidingTilesEnums.hpp](#)
[Main.cpp](#)
[UserFunctionsHelpers](#)

Definition in file [UserFunctions.cpp](#).

8.22 UserFunctions.cpp

[Go to the documentation of this file.](#)

```

00001
00019 #include "UserFunctions.hpp"
00020 #include "UserFunctionsHelpers.hpp"
00021 #include "SlidingTilesFunctions.hpp"
00022 #include "SlidingTilesData.hpp"
00023 #include "SlidingTilesEnums.hpp"
00024 #include <string>
00025 #include <iostream>
00026 #include <algorithm>
00027
00028 namespace UserFunctions {
00029     bool selectDifficulty() {
00030         //Infinite loop to ensure a valid option is picked
00031         while (true) {
00032             //Get and transform user input
00033             std::string input;
00034             std::cout << "SELECT: [Easy] [Medium] [Hard] [Insane] [Quit]\n" << std::endl;
00035             std::cin >> input;
00036             std::transform(input.begin(), input.end(), input.begin(),
00037                 [](unsigned char character) { return std::tolower(character); });
00038
00039             //Check if it is one of the options;
00040             if (input == "easy" || input == "e") {
00041                 SlidingTilesData::currentDifficulty = SlidingTilesEnums::Difficulty::EASY;
00042                 UserFunctionsHelpers::startGame();
00043                 return true;
00044             }
00045             else if (input == "medium" || input == "m") {
00046                 SlidingTilesData::currentDifficulty = SlidingTilesEnums::Difficulty::MEDIUM;
00047                 UserFunctionsHelpers::startGame();
00048                 return true;
00049             }
00050             else if (input == "hard" || input == "h") {
00051                 SlidingTilesData::currentDifficulty = SlidingTilesEnums::Difficulty::HARD;
00052                 UserFunctionsHelpers::startGame();
00053                 return true;
00054             }
00055             else if (input == "insane" || input == "i") {
00056                 SlidingTilesData::currentDifficulty = SlidingTilesEnums::Difficulty::INSANE;
00057                 UserFunctionsHelpers::startGame();
00058                 return true;
00059             }
00060             else if (input == "quit" || input == "q") {
00061                 return false;
00062             }
00063             else std::cout << "Invalid option" << std::endl;
00064         }
00065     }
00066
00067     bool performSlide() {
00068         //Infinite loop to ensure a valid option is picked
00069         while (true) {
00070             UserFunctionsHelpers::printBoard();
00071
00072             //Get and transform user input
00073             std::string input;
00074             std::cout << "SELECT: [LEFT/A] [RIGHT/D] [UP/W] [DOWN/S] [Quit]\n" << std::endl;
00075             std::cin >> input;
00076             std::transform(input.begin(), input.end(), input.begin(),
00077                 [](unsigned char character) { return std::tolower(character); });
00078
00079             //Check if it matches one of the options
00080             if (input == "left" || input == "a") {
00081                 UserFunctionsHelpers::trySlide(SlidingTilesEnums::Direction::LEFT);
00082                 return UserFunctionsHelpers::continueGame();
00083             }
00084             else if (input == "right" || input == "d") {
00085                 UserFunctionsHelpers::trySlide(SlidingTilesEnums::Direction::RIGHT);
00086                 return UserFunctionsHelpers::continueGame();
00087             }
00088             else if (input == "up" || input == "w") {
00089                 UserFunctionsHelpers::trySlide(SlidingTilesEnums::Direction::UP);
00090                 return UserFunctionsHelpers::continueGame();
00091             }
00092             else if (input == "down" || input == "s") {
00093                 UserFunctionsHelpers::trySlide(SlidingTilesEnums::Direction::DOWN);
00094                 return UserFunctionsHelpers::continueGame();
00095             }
00096             else if (input == "quit") {
00097                 return false;
00098             }
00099             else std::cout << "Invalid option" << std::endl;

```

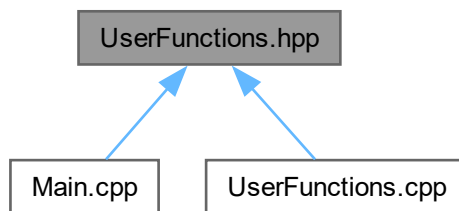


```
00100      }  
00101    }  
00102 }
```

8.23 UserFunctions.hpp File Reference

File containing a namespace with declared functions that the user interacts with.

This graph shows which files directly or indirectly include this file:



Namespaces

- namespace [UserFunctions](#)
Namespace containing functions that the user interacts with.

Functions

- bool [UserFunctions::selectDifficulty](#) ()
Allows the user to select their Sliding Tiles game difficulty.
- bool [UserFunctions::performSlide](#) ()
Allows the user to perform a slide in a Sliding Tiles game.

8.23.1 Detailed Description

This file contains a namespace with the declaration of functions that turn user input into output.

Author

Rajdeep Chowdhury

Date

8.04.2026

See also

[UserFunctions.cpp](#)

[Main.cpp](#)

[UserFunctionsHelpers](#)

Definition in file [UserFunctions.hpp](#).

8.24 UserFunctions.hpp

[Go to the documentation of this file.](#)

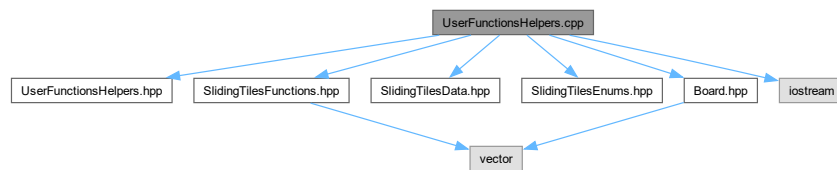
```
00001
00014 #pragma once
00015
00021 namespace UserFunctions {
00029     bool selectDifficulty();
00030
00040     bool performSlide();
00041 }
```

8.25 UserFunctionsHelpers.cpp File Reference

File containing the initialization of the [UserFunctionsHelpers](#) functions.

```
#include "UserFunctionsHelpers.hpp"
#include "SlidingTilesFunctions.hpp"
#include "SlidingTilesData.hpp"
#include "SlidingTilesEnums.hpp"
#include "Board.hpp"
#include <iostream>
```

Include dependency graph for UserFunctionsHelpers.cpp:



Namespaces

- namespace [UserFunctionsHelpers](#)

Namespace containing helper functions used by [UserFunctions](#).

Functions

- void [UserFunctionsHelpers::startGame](#) ()
Initializes data to start a game of Sliding Tiles.
- void [UserFunctionsHelpers::trySlide](#) ([SlidingTilesEnums::Direction](#) direction)
Tries to slide the empty tile in the given direction.
- void [UserFunctionsHelpers::printBoard](#) ()
Print the Sliding Tiles board.
- void [UserFunctionsHelpers::printLeaderboard](#) ()
Print the Sliding Tiles leaderboard for the current difficulty.
- bool [UserFunctionsHelpers::continueGame](#) ()
Determines whether the Sliding Tiles game should continue.

8.25.1 Detailed Description

This file contains the initialization of the [UserFunctionsHelpers](#) helper functions declared in [UserFunctionsHelpers.hpp](#).

Author

Rajdeep Chowdhury

Date

8.04.2026

See also

[UserFunctionsHelpers.hpp](#)
[SlidingTilesFunctions.hpp](#)
[SlidingTilesData.hpp](#)
[SlidingTilesEnums.hpp](#)
[Board.hpp](#)
[UserFunctions](#)

Definition in file [UserFunctionsHelpers.cpp](#).

8.26 UserFunctionsHelpers.cpp

[Go to the documentation of this file.](#)

```
00001
00018 #include "UserFunctionsHelpers.hpp"
00019 #include "SlidingTilesFunctions.hpp"
00020 #include "SlidingTilesData.hpp"
00021 #include "SlidingTilesEnums.hpp"
00022 #include "Board.hpp"
00023 #include <iostream>
00024
00025 namespace UserFunctionsHelpers {
00026     void startGame() {
00027         printLeaderboard();
00028         SlidingTilesFunctions::startGame(SlidingTilesData::currentDifficulty);
00029         SlidingTilesFunctions::shuffle();
00030     }
00031
00032     void trySlide(SlidingTilesEnums::Direction direction) {
00033         //Perform different results depending on if the slide succeeds or not
00034         bool success = SlidingTilesFunctions::slide(direction);
00035         if (success) SlidingTilesData::slides++;
00036         else std::cout << "No tile found in that direction." << std::endl;
00037     }
00038
00039     void printBoard() {
00040         //Use a nested for loop
00041         for (int i = 0; i < SlidingTilesData::boardSize; i++) {
00042             if (!SlidingTilesData::board.canAccess(i)) break; //For safety
00043             for (size_t tile : SlidingTilesData::board.access(i)) {
00044                 if (tile == SlidingTilesData::boardSize * SlidingTilesData::boardSize) std::cout << "[
00045 ]";
00046                 else if (tile <= 9) std::cout << "[ " << tile << ']'';
00047                 else std::cout << '[' << tile << ']'';
00048             }
00049             std::cout << '\n';
00050         }
00051
00052     void printLeaderboard() {
00053         //Load top scores then print them
```

```

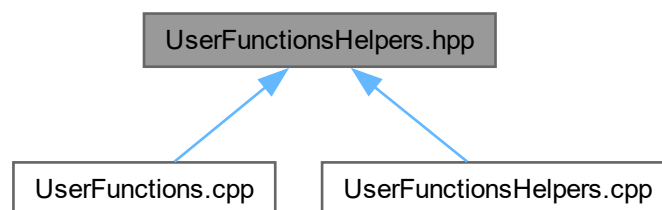
00054         std::vector<unsigned int> scores =
    SlidingTilesFunctions::loadScores(SlidingTilesData::currentDifficulty);
00055         std::cout << "\nLEADERBOARD\n";
00056         for (int i = 0; i < scores.size(); i++) {
00057             std::cout << i + 1 << ". " << scores.at(i) << '\n';
00058         }
00059         std::cout << '\n';
00060     }
00061
00062     bool continueGame() {
00063         using namespace SlidingTilesData;
00064         //The last move of the sliding tiles game is always making the empty tile go to the bottom
    right corner
00065         if (board.access(boardSize - 1, boardSize - 1) != boardSize * boardSize) return true;
00066         //Then check if the board is ordered to trigger a win
00067         else if (SlidingTilesFunctions::isBoardOrdered()) {
00068             SlidingTilesFunctions::writeScore(currentDifficulty);
00069             printBoard();
00070             std::cout << "Winner! \n";
00071             printLeaderboard();
00072             return false;
00073         }
00074         else return true;
00075     }
00076
00077 }

```

8.27 UserFunctionsHelpers.hpp File Reference

File containing a namespace of declared helper functions and a constant expression for [UserFunctions](#).

This graph shows which files directly or indirectly include this file:



Namespaces

- namespace [SlidingTilesEnums](#)
Namespace containing enum classes for a Sliding Tiles game.
- namespace [UserFunctionsHelpers](#)
Namespace containing helper functions used by [UserFunctions](#).

Functions

- void [UserFunctionsHelpers::startGame](#) ()
Initializes data to start a game of Sliding Tiles.
- void [UserFunctionsHelpers::trySlide](#) ([SlidingTilesEnums::Direction](#) direction)

- Tries to slide the empty tile in the given direction.*
 - void [UserFunctionsHelpers::printBoard](#) ()
Print the Sliding Tiles board.
 - void [UserFunctionsHelpers::printLeaderboard](#) ()
Print the Sliding Tiles leaderboard for the current difficulty.
 - bool [UserFunctionsHelpers::continueGame](#) ()
Determines whether the Sliding Tiles game should continue.

8.27.1 Detailed Description

This file contains a namespace of declared helper functions that the functions of [UserFunctions](#) use.

Author

Rajdeep Chowdhury

Date

8.04.2026

See also

[UserFunctionsHelpers.cpp](#)

[UserFunctions](#)

Definition in file [UserFunctionsHelpers.hpp](#).

8.28 UserFunctionsHelpers.hpp

[Go to the documentation of this file.](#)

```
00001
00013 #pragma once
00014
00015 //Forward declaration
00016
00017 namespace SlidingTilesEnums {
00018     enum class Direction;
00019 }
00020
00027 namespace UserFunctionsHelpers {
00033     void startGame();
00034
00040     void trySlide(SlidingTilesEnums::Direction direction);
00041
00046     void printBoard();
00047
00056     void printLeaderboard();
00057
00065     bool continueGame();
00066 }
```

